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DATA SCIENCE

PROJECT 2

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Introduction

The chosen dataset was named Vaccination against Covid-19. This dataset contains information on COVID-19 vaccinations that are categorized by age groups. There are 332,424 rows, which refer to the number of records. There are 10 columns, which are StatisticsDate (date of the recorded statistic, like “2020-12-27”), TargetDiseaseCode (numeric code for the disease, like “219” for COVID-19), TargetDisease (name of the disease for COVID-19), AgeGroup (age group of the population), VaccinationSeries (vaccination stage or dose number), MeasurementType (type of measurement like “FullyVaccinated”), LocationPopulation (population size of the location), DailyCount (daily count of vaccination or related measurements), TotalCount (cumulative count of vaccinations or related measurement) and PopulationCoverage (coverage percentage of the population). This dataset captures vaccination over time, most likely from multiple locations or demographic groups. This dataset is structured to allow for investigating vaccination trends, effectiveness and coverage rates based on age and other criteria. It keeps track of demographic information, population coverage, daily and cumulative vaccination counts. This dataset was downloaded from the Kaggle website (https://www.kaggle.com/datasets/olaflundstrom/vaccination-against-covid-19?resource=download).

The objective of this analysis is to evaluate the vaccine trends over time by comparing the changes in daily and total vaccination rates throughout the pandemic. The percentage of the population that is vaccinated can be analyzed by breaking down the age groups and other demographic features. Next, this analysis can identify the age groups that vaccination programs prioritized and those received less attention. The distribution of different vaccination metrics can also be examined, such as doses administered versus those who received all recommended vaccinations. By having this information, there are some areas that can be improved by identifying the age groups or time periods with low vaccination rates or coverage to drive public health efforts.

The aim of this analysis is to provide practical information about the success and shortcomings of vaccination programs. Next, assist policymakers in targeting under-vaccinated populations. To raise the awareness of vaccination, there are some of the demographic groups that need to be highlighted in order to encourage vaccination adoption. Lastly, this data can be used to provide guidance for next vaccination campaigns or pandemic responses.

Data Cleaning

pandas and os are two Python libraries that were imported in the Jupyter Notebook for data manipulation and analysis. The pd.read_csv() function was used to import the dataset (opendata_covid19_vaccination_agegroup.csv) into a Pandas DataFrame. This ensures that all the data is loaded into a manageable format.

```
summary_missing=data.isna().sum()
print(summary_missing)
```

```
StatisticsDate              0
TargetDiseaseCode           0
TargetDisease               0
AgeGroup                27702
VaccinationSeries           0
MeasurementType             0
LocationPopulation      27702
DailyCount                  0
TotalCount                  0
PopulationCoverage     172044
dtype: int64
```

Figure 1.1: The input and output for checking missing data

The missing data in the dataset was checked using the above code. This code summarizes the count of missing data in each column of the dataset. Based on the figure above, there are three columns that contain missing data, which are AgeGroup, LocationPopulation and Population Coverage.

AgeGroup and LocationPopulation columns look very important in the analysis. Incomplete or invalid records drawn from both columns could cause misleading information. Without AgeGroup, it is impossible to proceed with the analysis without having the age range data. Per capita metrics and percentage coverage calculations would be inaccurate without LocationPopulation data.

```
#Handling missing data
data=data.dropna(subset=['AgeGroup','LocationPopulation'],axis=0)
data['PopulationCoverage']=data['PopulationCoverage'].fillna(0)
```

Figure 1.2: The input for handling the missing data

Therefore, rows with missing data in AgeGroup and LocationPopulation columns were dropped. By removing missing data in AgeGroup and LocationPopulation, the remaining dataset is guaranteed to be valid and reliable for the analysis. Next, missing data in the PopulationCoverage column were replaced with 0. PopulationCoverage is a derived statistic and is not a necessary input and the missing data was replaced with 0. The records of missing data are probably those where vaccination data has not yet been recorded or computed. It makes sense to assume this coverage is 0 in this situation since it prevents the unnecessary removal of huge datasets. Filling the missing data with 0 avoids problems during computations or visualizations.

```
#Recheck missing data
summary2_missing=data.isna().sum()
print(summary2_missing)
```

```
StatisticsDate        0
TargetDiseaseCode     0
TargetDisease         0
AgeGroup              0
VaccinationSeries     0
MeasurementType       0
LocationPopulation    0
DailyCount            0
TotalCount            0
PopulationCoverage    0
dtype: int64
```

Figure 1.3: The input and output to recheck the missing data

The code above shows the step of rechecking the missing data after removing and replacing it. This step is used to verify that the operation was successful.

```
#check duplicated rows
dupli=data.duplicated().sum()
print(dupli)
```

```
0
```

Figure 1.4: The input and output to check the duplicate rows

The code in Figure 1.4 was used to check the duplicate rows in the dataset. The output will print the total number of duplicated rows. In this dataset, there are no duplicate rows since the output is 0.

```python
for column_name in data.columns:
    print(f'Column:{column_name}')
    unique_value=data[column_name].unique()
    print(unique_value)
```

Figure 1.5: The input to inspect unique values in each column

```
Column:StatisticsDate
['2020-12-27' '2020-12-28' '2020-12-29' ... '2024-12-21' '2024-12-22'
 '2024-12-23']
Column:TargetDiseaseCode
[219]
Column:TargetDisease
['COVID-19']
Column:AgeGroup
['0-4' '12-15' '16-17' '18-29' '30-39' '40-49' '5-11' '50-59' '60-69'
 '70-79' '80+']
Column:VaccinationSeries
[ 1  2  3  4  5  6  7  8 10]
Column:MeasurementType
['DosesAdministered' 'FullyVaccinated' 'Vaccinated']
Column:LocationPopulation
[ 71235.  56876.  24982. 169308. 196352. 182795. 103951. 171453. 163237
 111342.  77358.  70321.  59238.  25917. 162764. 198484. 183010. 102751
 171103. 164770. 112691.  79019.  69534.  60788.  27186. 159545. 199035
 183636. 101830. 171562. 165599. 112898.  80183.  69643.  64893.  29335
 162786. 205829. 189109. 105240. 174783. 168239. 115840.  80187.]
Column:DailyCount
[  0  11  40 ... 774 582 750]
Column:TotalCount
[    0    11    40 ... 12114 12159  5397]
Column:PopulationCoverage
[0.000e+00 1.000e-02 2.000e-02 ... 1.411e+01 6.710e+00 1.516e+01]
```

Figure 1.6: The output for unique values in each column

The unique values in each column of the dataset are examined using the above code. This code assists in understanding the structure of the data, identifying inconsistencies and analyzing for potential errors.

C	D	
etDise	AgeGroup	Va
ID-19	0-4	
ID-19	0-4	
ID-19	0-4	
ID-19	15-Dec	
ID-19	15-Dec	
ID-19	15-Dec	
ID-19	16-17	
ID-19	16-17	
ID-19	16-17	
ID-19	11-May	
ID-19	11-May	
ID-19	11-May	

Figure 1.7: Values for AgeGroup column in excel

```python
#Check unique values
for column_name in data.columns:
    print(f'Column:{column_name}')
    unique_value=data[column_name].unique()
    print(unique_value)
```

```
Column:StatisticsDate
['2020-12-27' '2020-12-28' '2020-12-29' ... '2024-12-21' '2024-12-22'
 '2024-12-23']
Column:TargetDiseaseCode
[219]
Column:TargetDisease
['COVID-19']
Column:AgeGroup
['0-4' '12-15' '16-17' '18-29' '30-39' '40-49' '5-11' '50-59' '60-69'
 '70-79' '80+']
Column:VaccinationSeries
[ 1  2  3  4  5  6  7  8 10]
Column:MeasurementType
['DosesAdministered' 'FullyVaccinated' 'Vaccinated']
```

Figure 1.8: Values for AgeGroup column in Python

The dataset in Excel (Figure 1.7) shows incorrect or misinterpreted values for 15-Dec and 11-May, but the unique() function in Python (Figure 1.8) shows the correct values for the AgeGroup column for 12-15 and 5-11. The problem arises because Excel automatically converts text values that match the dates into actual date formats. When checked in Python using the unique() function, the proper values are displayed since Python preserves the data in its original format without automatic transformations. This confirms that the dataset itself was correctly formatted and the problem only occurred when viewed in Excel.

```
#Check data types
data.dtypes
```

```
StatisticsDate          object
TargetDiseaseCode        int64
TargetDisease           object
AgeGroup                object
VaccinationSeries        int64
MeasurementType         object
LocationPopulation     float64
DailyCount               int64
TotalCount               int64
PopulationCoverage     float64
dtype: object
```

Figure 1.9: The input and output for data types of each column

The figure above shows the data types of each column. This step is important, especially when performing data analysis or preprocessing. Checking data types ensures that the data is clean, accurate and ready for analysis. This step also ensures the data can be processed in the right way. Thus, the processing step can be done without any errors or discrepancies. The StatisticsDate is a date column, but the data type of this column shows the object type, which indicates that the dates in this column have not yet been correctly transformed to date time. It must be converted to datetime format in order to be used for time-based analysis.

The TargetDiseaseCode is an int64 type. This column contains numeric codes like 2, 1 and 9. It is stored as a 64-bit integer, which is standard form for numeric data. TargetDisease is a categorical column with text descriptions of target disease (COVID-19). In pandas, objects representing string data are utilized for textual information. The AgeGroup column contains age group labels (‘12-15,’ ‘16-17’) and is stored as objects, which is standard for string-based categorical columns.

The VaccinationSeries column is int64 type. This column provides numeric data that represent the vaccination series, such as ‘1’ for the first dose and ‘2’ for the second dose. The MeasurementType is an object type. This column provides descriptive names for the type of measurement, such as fully vaccinated and doses administered.

The LocationPopulation and PopulationCoverage are float64 types. Both columns provide numeric data (possibly with decimals) that represent the populations of various locations. The float64 type is appropriate for storing continuous numeric data with decimal points. The DailyCount and TotalCount are int64 types. This column contains integer values representing the daily counts of vaccinations or related statistics.

```
#Convert to datetime format
data['StatisticsDate']=pd.to_datetime(data['StatisticsDate'],errors='coerce')
data.head()
```

	StatisticsDate	TargetDiseaseCode	TargetDisease	AgeGroup	VaccinationSeries	Mea
19	2020-12-27	219	COVID-19	0-4	1	Dos
20	2020-12-27	219	COVID-19	0-4	1	
21	2020-12-27	219	COVID-19	0-4	1	

Figure 1.10: The input and output for converting the datetime format

Figure 1.10 shows the step of converting the StatisticsDate column to datetime type. This step helps in making the operation easier for filtering, sorting or analyzing.

```
#Recheck data types
data.dtypes
```

```
StatisticsDate          datetime64[ns]
TargetDiseaseCode                int64
TargetDisease                   object
AgeGroup                      category
VaccinationSeries                int64
MeasurementType                 object
LocationPopulation            category
DailyCount                       int64
TotalCount                       int64
PopulationCoverage            category
dtype: object
```

Figure 1.11: The input and output for rechecking the data types of each column

Rechecking the data types of each column in the dataset is an important step in the data cleaning and preprocessing workflow. It ensures that the data is well-formatted and suitable for analysis and modeling. Figure 1.11 shows that the data types are in the correct data types after some changes in the previous step.

```
#Standardize categorical values
data['MeasurementType']=data['MeasurementType'].str.replace(r'([a-z])([A-Z])',r'\1 \2',regex=True)
print(data['MeasurementType'].unique())
```

```
['Doses Administered' 'Fully Vaccinated' 'Vaccinated']
```

Figure 1.12: Standardize the categorical column

To correct the inconsistencies in naming conventions, the space was added between a lowercase letter ([a-z]) and an uppercase letter ([A-Z]) in the MeasurementType column. This increases readability and guarantees consistency in formatting of the text.

```
#Standardize column names
data.rename(columns={
    'StatisticsDate':'Statistics Date',
    'TargetDiseaseCode':'Target Disease Code',
    'TargetDisease':'Target Disease',
    'AgeGroup':'Age Group',
    'VaccinationSeries':'Vaccination Series',
    'MeasurementType':'Measurement Type',
    'LocationPopulation':'Location Population',
    'DailyCount':'Daily Count',
    'TotalCount':'Total Count',
    'PopulationCoverage':'Population Coverage',
},inplace=True)

print(data.columns)
```

```
Index(['Statistics Date', 'Target Disease Code', 'Target Disease', 'Age Group',
       'Vaccination Series', 'Measurement Type', 'Location Population',
       'Daily Count', 'Total Count', 'Population Coverage'],
      dtype='object')
```

Figure 1.13: Standardize the column names

The code snippet standardizes the dataset's column names by renaming them in a consistent way. When the code runs correctly, it changes the column names in the DataFrame to the new standardized ones.

Analysis and Insights

```python
#Extract descriptive statistics
descriptive_stats=data.describe(include='all')
print(descriptive_stats)
```

```
            Statistics Date  Target Disease Code Target Disease Age Group  \
count                304722             304722.0         304722    304722
unique                  NaN                  NaN              1        11
top                     NaN                  NaN       COVID-19       0-4
freq                    NaN                  NaN         304722     27702
mean    2022-12-25 12:00:00                219.0            NaN       NaN
min     2020-12-27 00:00:00                219.0            NaN       NaN
25%     2021-12-26 00:00:00                219.0            NaN       NaN
50%     2022-12-25 12:00:00                219.0            NaN       NaN
75%     2023-12-25 00:00:00                219.0            NaN       NaN
max     2024-12-23 00:00:00                219.0            NaN       NaN
std                     NaN                  0.0            NaN       NaN


        Vaccination Series   Measurement Type  Location Population  \
count        304722.000000             304722        304722.000000
unique                 NaN                  3                  NaN
top                    NaN  Doses Administered                 NaN
freq                   NaN             144342                  NaN
mean              4.894737                NaN        122568.832585
min               1.000000                NaN         24982.000000
25%               2.000000                NaN         70321.000000
50%               5.000000                NaN        112898.000000
75%               7.000000                NaN        171103.000000
max              10.000000                NaN        205829.000000
std               2.845032                NaN         55672.289595

          Daily Count    Total Count  Population Coverage
count   304722.000000  304722.000000        304722.000000
unique            NaN            NaN                  NaN
top               NaN            NaN                  NaN
freq              NaN            NaN                  NaN
mean        15.255242   17958.520497             6.897090
min          0.000000       0.000000             0.000000
25%          0.000000       0.000000             0.000000
50%          0.000000       0.000000             0.000000
75%          0.000000    2249.000000             0.000000
max       4859.000000  239244.000000            96.140000
std        113.561597   44978.474640            20.308158
```

Figure 2.1: The input for extract descriptive statistics

Figure 2.1 shows the summary statistics for all columns in the dataset. The summary statistics include count, unique values, top (most frequent value), frequency, mean, minimum, 25%, 50%, 75%, maximum and standard deviation for the columns.

```
#Extract the mode for each column
mode_column=data.mode().iloc[0]
print(mode_column)
```

```
Statistics Date          2020-12-27 00:00:00
Target Disease Code                    219.0
Target Disease                      COVID-19
Age Group                                0-4
Vaccination Series                       1.0
Measurement Type          Doses Administered
Location Population                  29335.0
Daily Count                              0.0
Total Count                              0.0
Population Coverage                      0.0
Name: 0, dtype: object
```

Figure 2.2: The mode for each column

Figure 2.2 shows the code extracting a single row with the most frequent value for each column. iloc[0] will select the first row of the result from data.mode(). This ensures that only one mode value is selected if there are multiple modes for a column.

```
#Correlation analysis (for numerical variables)
numerical=data.select_dtypes(include=['number'])
correlation=numerical.corr()
print(correlation)
```

```
                     Target Disease Code  Vaccination Series  \
Target Disease Code                  NaN                 NaN
Vaccination Series                   NaN        1.000000e+00
Location Population                  NaN       -1.738864e-16
Daily Count                          NaN       -1.633732e-01
Total Count                          NaN       -5.052418e-01
Population Coverage                  NaN       -4.313147e-01

                     Location Population  Daily Count  Total Count  \
Target Disease Code                  NaN          NaN          NaN
Vaccination Series         -1.738864e-16    -0.163373    -0.505242
Location Population         1.000000e+00     0.072684     0.236456
Daily Count                 7.268424e-02     1.000000     0.137733
Total Count                 2.364561e-01     0.137733     1.000000
Population Coverage         8.561317e-02     0.070122     0.546298


                     Population Coverage
Target Disease Code                  NaN
Vaccination Series             -0.431315
Location Population             0.085613
Daily Count                     0.070122
Total Count                     0.546298
Population Coverage             1.000000
```

Figure 2.3: The input and output for correlation analysis

Output Figure 2.3 shows the measure of the linear relationship between numerical variables in the dataset. The correlation of Vaccination Series vs Total Count is -0.5052. These results indicate a moderate negative correlation, which means that the total count of vaccinations tends to decline as the Vaccination Series increases. This could reflect that some Vaccination Series have lower associated counts, possibly due lower uptake or targeted campaigns.

The correlation between Location Population vs Total Count is 0.2365. There is a weak positive correlation observed between them. Larger populations tend to have higher total vaccination counts, but the relationship is not particularly strong. This implies that there are more factors that play a role, like vaccination policies or availability.

The correlation between Population Coverage vs Total Count is 0.5463. Areas with greater total vaccination counts also will have better population coverage. This illustrates the effectiveness of vaccination campaigns in increasing coverage as more doses are administered.

The correlation between Daily Count vs Total Count is 0.1377. A weak positive correlation suggests that daily vaccination rates slightly contribute to the overall vaccination totals. This weak link arises if the dataset contains many zero or low daily counts.

The correlation between Vaccination Series vs Population Coverage is -0.4313. A moderate negative correlation implies that Population Coverage may decrease as the Vaccination Series increases. It may also point to difficulties maintaining public interest in following vaccination schedules or possible outreach gaps for higher-dose programs.

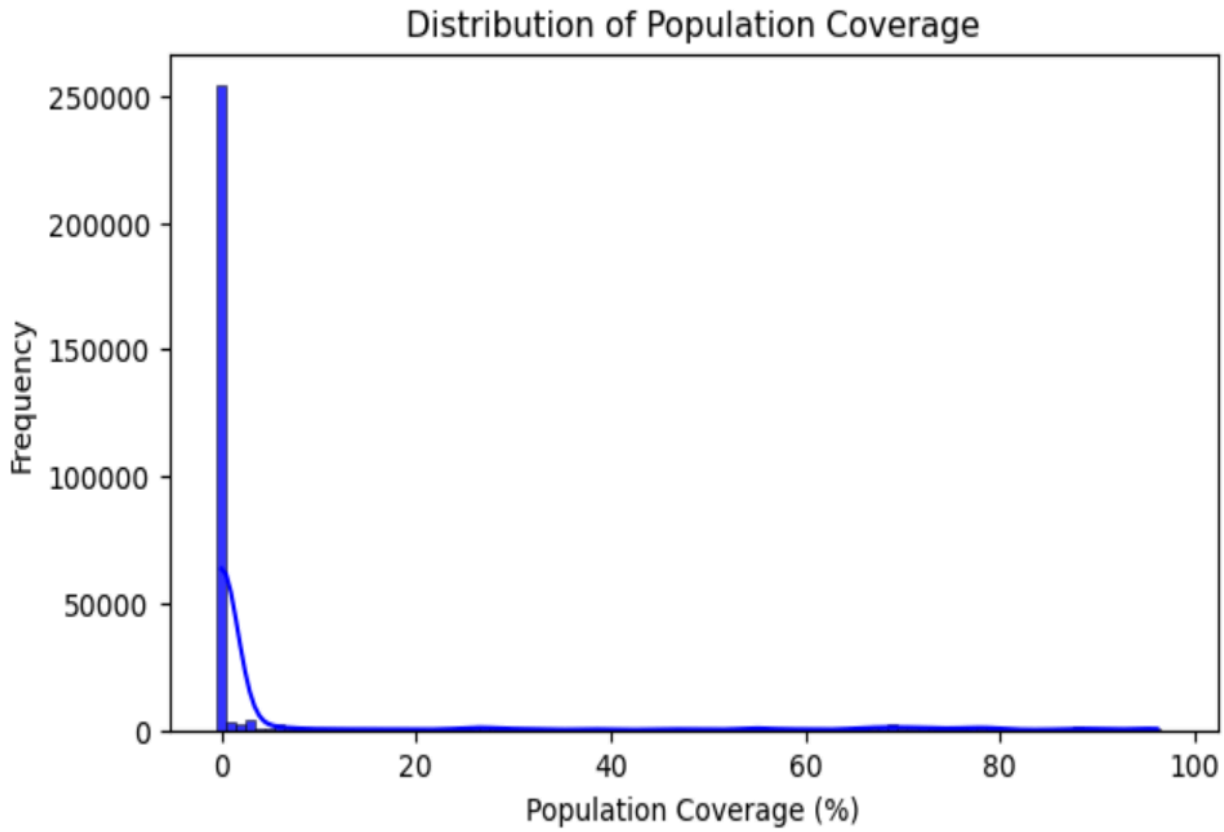

Figure 3.1: Distribution of Population Coverage

The distribution of Population Coverage above is positively skewed to the right since it has a longer whisker on the right side. Most of the values concentrated near 0%. The values on the right side of the histogram represent larger percentages of population coverage and are extremely infrequent. This indicates that only a small number of the population have received widespread vaccinations covering a large proportion of the population. The peak on the left side of the histogram (around 0% population coverage) shows that most regions or groups have hardly started or progressed in vaccinating their population. This difference highlights the possibility that vaccination campaigns were not spread equally and certain groups have received more focus than others. This could be a result of technical difficulties, tactics for setting priorities (such as focusing on specific groups or regions first), or other socio-political elements.

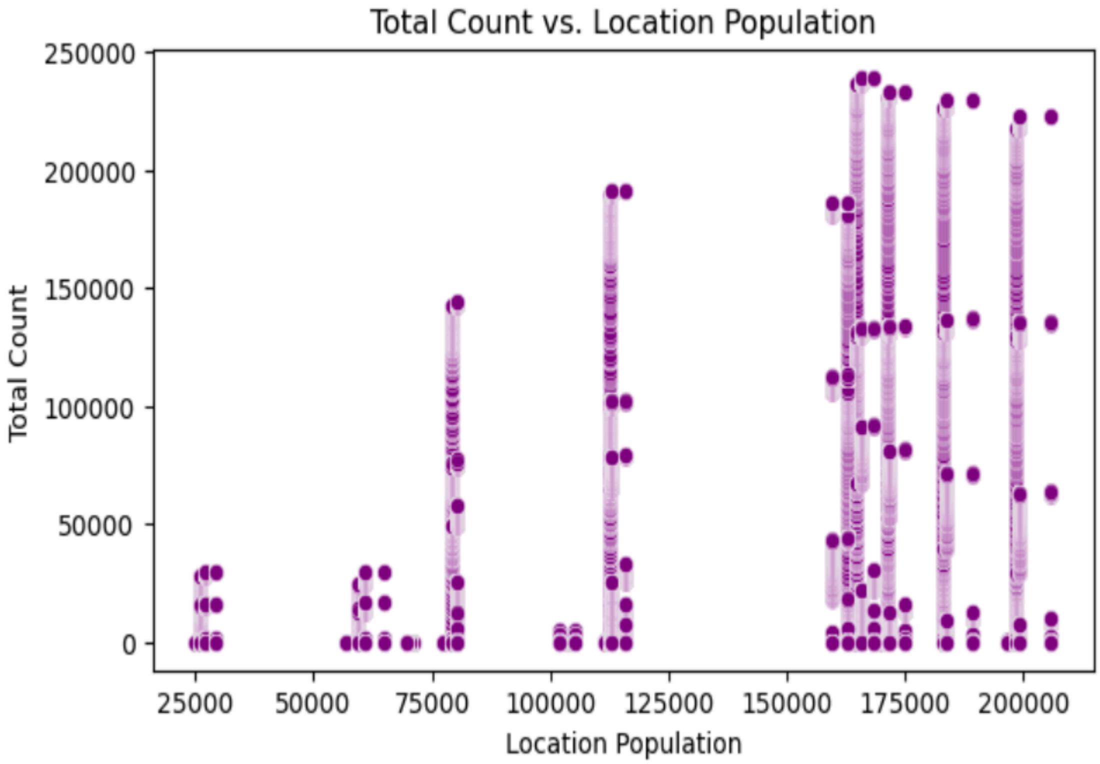

Figure 3.2: Total Count vs Location Population

The scatter plot illustrates the relationship between Location Population and Total Count. At specific population values, the data points cluster vertically. This suggests that the populations are dispersed according to distinct, predetermined sizes, which most likely correspond to standard demographic groupings or predetermined geographical classifications. Higher total counts are typically associated with larger populations. Total Count values are often greater with higher populations. This suggests that vaccination totals are positively correlated with population size. It makes sense that population size and vaccination rates would correlate and this relationship is crucial for determining how well resources are allocated in respect to population size. Data points from larger populations are more concentrated, which suggests that vaccination programs give them more attention. More scattered data points are seen in lower populations, suggesting that there are fewer such regions in the dataset. This visualization emphasizes the need to ensure that vaccine distribution strategies are aligned more effectively with regional population size.

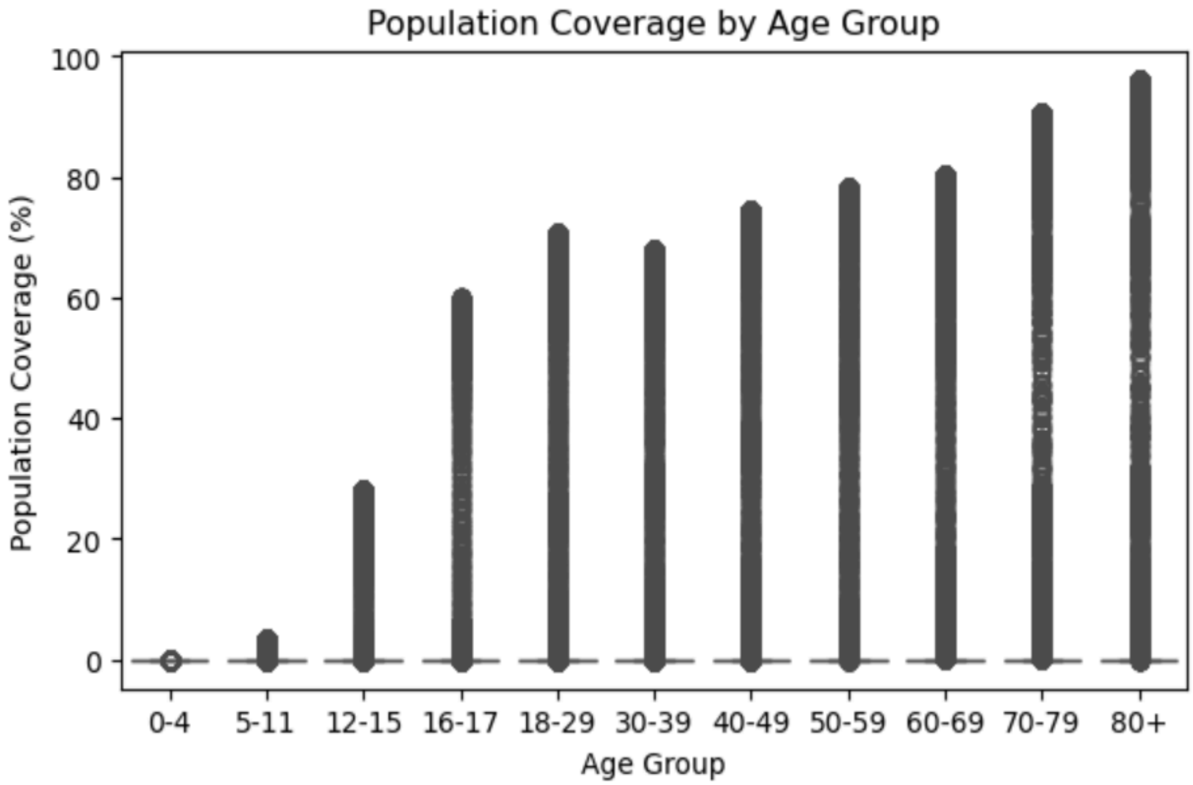

Figure 3.3: Population Coverage vs Age Group

This plot shows the Population Coverage (%) across different Age Groups. The youngest groups (0-15) have very little coverage. This might suggest that a certain age group has restricted access to eligibility for or uptake of vaccines. Teenage and adult groups in the middle groups (16-59) show a large rise in coverage, indicating higher vaccination or higher eligibility. The older group (70-80+) has the highest coverage levels. This pattern is consistent with vaccination prioritizing plans, which generally target older age groups. Potential vaccination priority strategies aimed at older and more vulnerable populations are reflected in the statistics. Low coverage among the youngest groups may be a result of vaccination hesitation, policy decisions or the vulnerability of pediatric vaccinations.

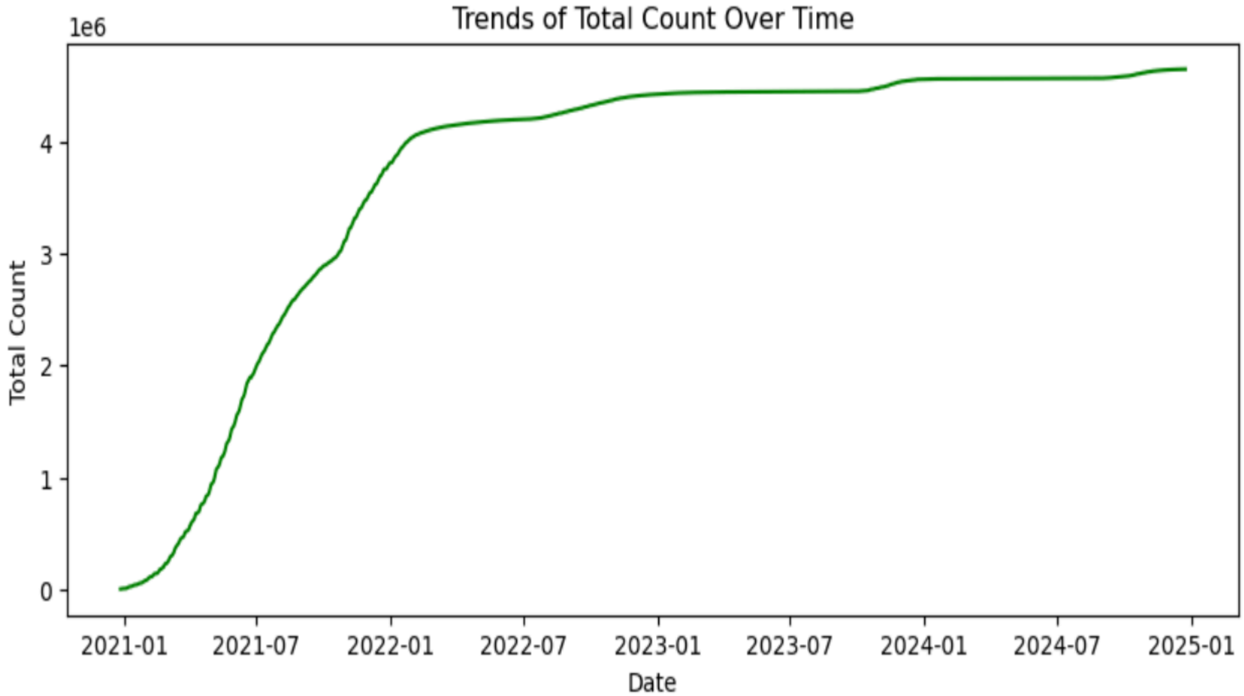

Figure 3.4: Trends of Total Count over Time

This line chart illustrates the cumulative trend of Total Counts over time, indicating how the total count of a particular measure (such as vaccination, cases or doses administered) has changed over time. The graph show that the upward trend indicates that the overall count has been steadily rising over time. This probably represents the cumulative progress made in tracking patients or delivering vaccinations. The slope gets less steep at specific times (after 2022), suggesting a declining pace of increase. Seasonal variables, decreased case reporting, or vaccination saturation may be the cause of this. The leveling off approaching 2025 suggests that the efforts are stabilizing or that most of the target population is covered. The upward trend indicates successful attempts to raise the number, like expanding data gathering or vaccination campaigns. Policymakers may decide to change their approach in response to the slowing growth rate, such as focusing on particular groups or removing obstacles to access. The timeline makes it easy to see when the most advancements were made, which makes it possible to analyze significant events or policies that took place during those times.

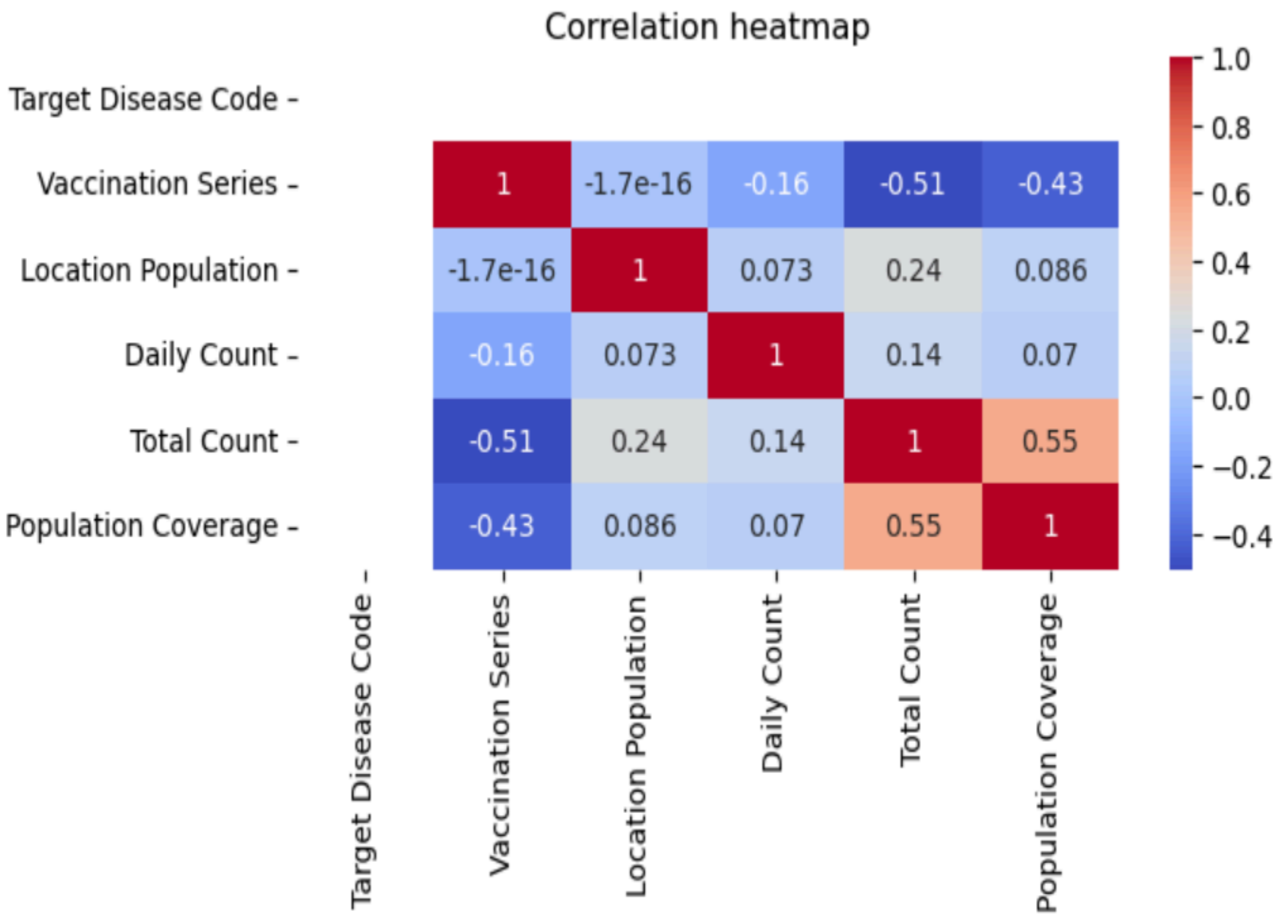

Figure 3.5: Correlation Heatmap

Figure 3.5 visualizes the relationship between different variables, where the values range from -1 to 1. Population Coverage and Total Count have the strongest positive correlation (0.55). This indicates that population coverage tends to rise in tandem with an increase in the total count. It shows the proportion of the population covered and the number of vaccinations given are directly correlated. Total Count and Vaccination Series have a correlation of 0.24. There may be a modest tendency for greater vaccination series values to correspond with higher total counts, as indicated by a weak positive correlation.

Target Disease Code and Total Count have a correlation of -0.51. This negative correlation may indicate that certain diseases are linked to lower total vaccination counts. A moderately negative correlation between Vaccination Series and Population Coverage (-0.43) suggests that greater vaccination series may be linked to lower population coverage as a result of the need for additional doses to achieve full coverage.

There is minimal to no linear relationship between variables such as Location Population and Daily Count, which exhibit nearly zero correlations. Effective vaccination efforts can increase overall coverage, as Population Coverage tends to increase together with Total Count. In the heatmap, a perfect positive correlation is represented by the number 1.0 (red). This indicates that there is a straight proportionality between the two variables and their changes are identical. Since the Target Disease Code's correlations with other variables are either insignificant or irrelevant (around zero), the heatmap produces no useful information about it. In this dataset, the Target Disease Code doesn't appear to offer any meaningful trends.

Conclusion

This dataset provides a depth analysis of the progress and difficulties of the COVID-19 vaccination campaign. It shows significant achievements in certain areas and among older age groups, where vaccination coverage is noticeably higher. It also highlights enduring inequalities, especially between younger populations and areas with lower vaccination rates. These results underline the need for targeted strategies to eliminate vaccination gaps and guarantee fair access for all groups. Policymakers, public health officials and medical professionals can use the conclusion from this analysis to have better vaccination programs, allocate resources more efficiently and enhance population safety and health in general.